

Junhui Chen

Ph.D. · Geoscientist — Coastal Geohazards & Geospatial Data Science

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PROFILE

Geoscientist (Ph.D., Chinese Academy of Sciences; CSC-funded joint training at Texas A&M University) working at the interface of marine / island spatial planning and quantitative geohazard science. I integrate remote sensing, InSAR surface-deformation monitoring, and machine learning to assess coastal and island hazards and to support territorial spatial planning and disaster risk reduction. 17 peer-reviewed publications (9 first-author SCI), four first-author manuscripts under review (one sole-authored), and 3 grants as Principal Investigator. Open to research and professional opportunities abroad — postdoctoral, research-scientist, and applied R&D roles — and to research collaboration.

RESEARCH INTERESTS

Coastal & marine geohazards; InSAR deformation monitoring (D-/PS-/SBAS-/DS-InSAR); island & territorial spatial planning; geospatial machine learning (susceptibility mapping, DEM error analysis); remote sensing; rock mechanics & microfracture.

EDUCATION

Ph.D., Cartography & Geographic Information Systems 2015 – 2020

Institute of Geographic Sciences and Natural Resources Research (IGSNRR), Chinese Academy of Sciences — Beijing

Joint-training Ph.D. (China Scholarship Council), Texas A&M University, Texas, USA (2019 – 2020)

B.Sc., Geology 2011 – 2015

China University of Geosciences (Beijing)

PROFESSIONAL EXPERIENCE

Assistant Researcher → Associate Researcher 2020 – present

Island Research Center, Ministry of Natural Resources — Fujian, China

- Territorial spatial planning for uninhabited islands at provincial (dominant-function zoning of 2,000+ islands, adopted into provincial plans), municipal, and single-island levels.
- Coastal geohazard & deformation research: susceptibility / risk assessment via remote sensing and machine learning; D-/PS-/SBAS-InSAR monitoring of landslides and reclamation areas; post-eruption deformation analysis of Tongatapu (2022).
- Machine-learning shipwreck susceptibility mapping for Chinese seas (information value, weights of evidence, neural networks, SVM, random forest).
- Policy & technical support: contributions to island-protection legislation, pilot schemes for legacy issues of uninhabited islands, and national “Beautiful Islands” technical review.

Secondment — Sea Area & Island Management Division 2023.7 – 2024.1

Municipal Bureau of Natural Resources and Planning, Xiamen

- Oversight of new-project sea use, mainland natural shoreline, and legacy reclamation issues; pre-research on transferring sea-area use rights via tender / auction / listing.

SELECTED PUBLICATIONS

First author unless noted; † under review. Full list on [Google Scholar](#).

1. † **Chen J.** Persistent subsidence of newly reclaimed coastal land: two-track InSAR evidence for a ground-readiness gap in land-use governance. **(sole author, under review)**

2. † **Chen J**, Tang F, Lin H, Huang B, Lin X. Post-Eruptive Crustal Relaxation and Stability of Tongatapu Island (2022–2025): Insights from Physics-Based InSAR Clustering. **(under review)**
3. † **Chen J**, Li S. From Resilience Debt to Regime Shift: The Synergistic Collapse of an Island's Eco-geomorphology (1987–2026). **(under review)**
4. † **Chen J**, Tang F, Lin H. Model-ready shipwreck susceptibility dataset for Chinese seas. **(under review)**
5. **Chen J**, Tang F, Lin H. Satellite-Derived Shorelines Reveal Typhoon-Driven Erosion and Monsoon-Gated Recovery on the Macrotidal Coast of Fujian, China. *Remote Sensing*, 2026, 18(17):3033. (IF 4.3)
6. **Chen J**, Tang F, Lin H. Coastal Vulnerability of the Macrotidal and Heavily Engineered Coast in Fujian, China. *Journal of Marine Science and Engineering*, 2026, 14(17):1654. (IF 3.2)
7. **Chen J**, Tang F, Lin H, Chen Y, Chen Y, Lin P, Huang B, Lin X. Geospatial predictive modelling of anthropogenic and natural shipwreck risks using multi-source environmental data. *npj Heritage Science*, 2026. (IF 3.9)
8. **Chen J**, Tang F, Lin H, Huang B, Lin X. Spatial Heterogeneity and Drivers of Vertical Error in Global DEMs: An Explainable Machine Learning Approach in Complex Subtropical Coastal Zones. *Remote Sensing*, 2026, 18(8):1125. (IF 4.1)
9. **Chen J**, Tang F, Lin H, Chen Y, Chen Y, Lin P, Huang B, Lin X. Application of Remote Sensing and Machine Learning Algorithms for Shipwreck Susceptibility Mapping in China. *Journal of Archaeological Science*, 2026, 185:106429. (TOP, IF 2.5)
10. **Chen J**, Tang F, Lin H, Huang B, Lin X. Mapping the research framework and key trends of coral reefs in the South China Sea. *Frontiers in Marine Science*, 2025, 12:1619275. (IF 3.0)
11. **Chen J**, Tang F. Visualized analysis of microscale rock mechanism research: A bibliometric data mining approach. *Heliyon*, 2024, 10(20):e39160. (IF 3.4)
12. **Chen J**, Tang F. Analyzing Contact Heterogeneity and Its Geological Implications in Granite Uniaxial Compression Tests. *Springer Series in Geomechanics and Geoengineering*, 2024. (EI)
13. **Chen J**, Feng A. The Effect of Thermal Discharging on the Island Ecosystem in China: Current Knowledge and Future Challenges. *Journal of Physics: Conference Series*, 2023, 2468(1). (EI)
14. **Chen J**, Lan H, Macciotta R, et al. Microfracture characterization of shale constrained by mineralogy and bedding. *Journal of Petroleum Science and Engineering*, 2021, 108456. (TOP, IF 4.3)
15. Lan H, **Chen J**, Macciotta R. Universal confined tensile strength of intact rock. *Scientific Reports*, 2019, 9(1):6170. (IF 4.0)
16. **Chen J**, Lan H, Macciotta R, et al. Anisotropy rather than transverse isotropy in Longmaxi shale and the potential role of tectonic stress. *Engineering Geology*, 2018, 247:38–47. (IF 3.9)
17. Zhang Y, Lan H, Li L, Wu Y, **Chen J**, Tian N. Optimizing the frequency ratio method for landslide susceptibility assessment: A case study of the Caiyuan Basin in the southeast mountainous area of China. *Journal of Mountain Science*, 2020, 17(2):340–357. (IF 2.5)

Plus four additional peer-reviewed articles published in Chinese (*Journal of Engineering Geology*; *Chinese Journal of Geophysics*).

FUNDED PROJECTS

Principal Investigator

- Refined assessment of geo-environmental stability in typical Fujian coastal zones based on multi-source remote sensing fusion — **Fujian Natural Science Foundation, General Program** (2026–2029)
- Compilation of guidelines for delineating underwater cultural heritage reserves — **National Cultural Heritage Administration** (2023–2024)
- Influence of mineral mechanical properties on fracture evolution in granite under uniaxial compression — **Fujian Natural Science Foundation, Youth Innovation Program** (2022–2025)

Key Participant

- Ecological protection services and decision support for sea areas and islands — **Ministry of Natural Resources** (2020–2022)
- Fundamental theories and key technologies for shale gas exploration and development — **CAS Strategic Priority Research Program** (2014–2018)

HONORS

- Fujian Provincial High-level Talent (2022); Pingtan Mid-/High-level Talent
- Keynote speaker, 12th Common Homeland Forum
- AGU Fall Meeting Student Grant (2017)

TECHNICAL SKILLS

GIS & Remote Sensing: ArcGIS, remote-sensing interpretation, GoCAD (3D geo-modeling)

InSAR: D-InSAR, PS-InSAR, SBAS-InSAR, DS-InSAR

Programming & Machine Learning: Python; random forest, SVM, neural networks

Rock-mechanics testing: DIC, uniaxial compression, UDEC

Languages: Chinese (native); English (academic — AGU / ISRM presentations, joint Ph.D. in the US)